



Nickel toxicity to benthic organisms: The role of dissolved organic carbon, suspended solids, and route of exposure



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ABSTRACT

Nickel bioavailability is reduced in the presence of dissolved organic carbon (DOC), suspended solids (TSS), and other complexing ligands; however, no studies have examined the relative importance of Ni exposure through different compartments (water, sediment, food). *Hyalella azteca* and *Lymnaea stagnalis* were exposed to Ni-amended water, sediment, and food, either separately or in combination. Both organisms experienced survival and growth effects in several Ni compartment tests. The DOC amendments attenuated *L. stagnalis* Ni effects (survival, growth, and ⁶²Ni bioaccumulation), and presence of TSS exposures demonstrated both protective and synergistic effects on *H. azteca* and *L. stagnalis*. ⁶²Ni trophic transfer from food to *H. azteca* and *L. stagnalis* was negligible; however, bioaccumulating ⁶²Ni was attributed to ⁶²Ni-water (⁶²Ni flux from food), ⁶²Ni-TSS, and ⁶²Ni-food. Overall, *H. azteca* and *L. stagnalis* Ni compartment toxicity increased in the following order: Ni-water >> Ni-sediment >> Ni-all (water, sediment, food) >> Ni-food.

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1. Introduction

Metals enter aquatic environments through both natural sources (rock weathering, volcanoes, biogenic processes), and anthropogenic sources (mining, smelting, fossil fuel combustion) (Callender, 2003; Sen Gupta and Bhattacharyya, 2008). Once these metals enter aquatic ecosystems, they can become bioavailable to organisms, and can cause toxicity (Callender, 2003). Nickel is one such metal, and its ecotoxicology is experiencing recent attention in the literature (Costello et al., 2011, 2012; Nguyen et al., 2011; Vangheluwe et al., 2013; Croteau and Luoma, 2008). Relating the relative aquatic toxicity of Ni to the major exposure compartments (water, sediment, food) studies has received limited attention in the scientific literature. However, authors have characterized Ni toxicity in single compartments (water, sediment, food), and these have ranged from survival to bioaccumulation effects (Watras et al., 1985; Nowierski et al., 2005; Doig and Liber, 2006; Meyer et al., 2007; Croteau and Luoma, 2008, 2009; Cloran et al., 2010).

There are several factors affecting the bioavailability Ni in water and sediments, and these include: dissolved organic carbon (DOC),

total suspended solids (TSS), acid volatile sulfides (AVS), Fe/Mn (oxy)hydroxides, and organic carbon (OC) present in water and sediments (Pyle et al., 2002; USEPA, 2005; Cloran et al., 2010). Metals, including Ni, have an affinity to these variables (DOC, TSS, OC, and AVS), and it is imperative to examine these variables for our understanding of Ni toxicity in water and sediments (USEPA, 2005; Sen Gupta and Bhattacharyya, 2008; Cloran et al., 2010). DOC is an important ligand for metals, and increased DOC concentrations have been shown to reduce metal bioavailability in waterborne exposures. However, metal toxicity is not always attenuated with increased DOC, and appears to be organism specific (De Schampelaere et al., 2004; Doig and Liber, 2007, 2006; Cloran et al., 2010).

Once metals are associated with food they can become a dietary route of exposure (Doig and Liber, 2006). When determining metal uptake by organisms, others (Watras et al., 1985; Courtney and Clements, 2002; Wilding and Maltby, 2006; Nguyen et al., 2012) have spiked metals with food to follow organism toxicity and bioaccumulation. Croteau and Luoma (2008, 2009) have shown that metals either adsorb or absorb onto food, and were readily accumulated by *Lymnaea stagnalis* during feeding tests. Croteau and Luoma (2008) showed that *L. stagnalis* tissue content was higher in Cd, Cu, and Ni during feeding rather than water-only acute metal exposures. Organismal metal bioaccumulation can have food chain

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and food web effects that potentially can alter ecological function of a system (Hare, 1992). The objectives of this study were to (1) discern *Hyalella azteca* and *L. stagnalis* toxicity in multiple compartments (water, sediment, food); (2) determine the role that DOC and TSS amendments have on Ni bioavailability in these multiple compartments; (3) and whether the stable isotope ^{62}Ni bioaccumulates during feeding in these multiple compartment exposures.

2. Methods

2.1. Laboratory experimental design

L. stagnalis and *H. azteca* were exposed to a series of Ni-amended compartments (Ni-water, Ni-sediments, Ni-food, Ni-all (water, sediment, food)), and each compartment received overlying water amendments of TSS (425 μm sieved WD sediment), DOC (Aldrich Humic acid), or none (SI, Fig. S1). Each organism was exposed simultaneously on two sediment types collected from: Greenville Creek (GC), Ohio, USA (low AVS and OC), and Warden Ditch (WD), Ohio, USA (high AVS and OC). The four Ni exposures were: (Ni-water) Ni-amended water + clean sediment + clean food, (Ni-sediment) clean water + Ni-amended sediments + clean food, (Ni-food) clean water + clean sediment + ^{62}Ni -amended food, (Ni-all) Ni-amended water + Ni-amended sediments + ^{62}Ni -amended food, and a (Control) clean water + clean sediment + clean food (SI, Fig. S1). Control treatments represent baseline *H. azteca* and *L. stagnalis* responses to TSS and DOC without Ni amendments (Table 1 and SI, Tables S1 and S2).

Each 7 d Ni exposure had four treatments (reference, TSS, DOC, and Ni-only/None), four replicates of 10 organisms, with one of these replicates/treatment used strictly for sediment and water chemistry analysis. *H. azteca* were 7–14 d old (USEPA, 2000) and *L. stagnalis* were <7 d post-hatch at the start of each test (Brix et al., 2011). Organisms were exposed in 300 ml high lip beakers with 100 ml of sediment, and ~175 ml of overlying water. Water changes (culture water, Ni-water, TSS-water, and DOC-water) were delivered using a Zumwalt design that delivered ~1 L of water dispersed over all beakers twice a day.

2.2. Ni concentrations for the Ni-exposures

Nickel concentrations for all the water, sediment, and food (compartments) exposures were held constant to allow for

comparisons of *L. stagnalis* and *H. azteca* Ni effects. However, due to varying sensitivities of *L. stagnalis* and *H. azteca*, the Ni-water concentrations required adjusting. Nickel LC₇₅ concentrations for water and sediments were chosen to discern Ni effects with different sediment types, Food, DOC, and TSS additions. *L. stagnalis* Ni-water concentrations were spiked at ~400 $\mu\text{g/L}$ of Ni, and *H. azteca* Ni-water concentrations at ~2000 $\mu\text{g/L}$ of Ni. In the Ni-sediment exposures, both WD sediments (WDsed) and GC sediments (GCsed) were spiked with similar Ni concentrations (~200 mg/kg of Ni) (SI, Tables S1 and S2). The leaf and lettuce disks used in the Ni-food exposures were soaked in 1200 $\mu\text{g/L}$ of ^{62}Ni . Changes were made to the Ni concentrations used during the Ni-all exposures since all compartments (water, sediment, and food) received Ni amendments. The Ni concentrations were reduced by half in the Ni-all exposures (water ~1200 $\mu\text{g/L}$ of Ni, sediment ~100 mg/kg of Ni, and food soaked in 600 $\mu\text{g/L}$ of ^{62}Ni) (SI, Tables S1 and S2).

2.3. DOC, and TSS amendments to water and sediments

The TSS-water pulse exposures were for 24 h, and after this time, the appropriate water source (culture-water, Ni-amended water) was given minus the TSS. The DOC-water and Ni-water was added for the duration of the exposure. The use of air pumps and glass pipette tips were used to suspend the TSS, and all treatments (Reference, TSS, DOC, and Ni-only) received airlines during the 24 h exposures.

Water samples for Ni and DOC were taken on Days 1 and 7. The Ni and DOC samples were filtered through acid cleaned 0.45 μm filters to determine dissolved Ni and OC fractions. All DOC concentrations were analyzed on Tekmar/Teledyne TOC combustion analyzer. The TSS samples were filtered through Whatman® filters for TSS determination, and separate water samples were analyzed for turbidity (NTU).

2.4. Food source and ^{62}Ni food labeling

L. stagnalis replicates were fed one romaine lettuce disk (17 mm in diameter), and *H. azteca* replicates were fed three microbial conditioned *Acer rubrum* leaf disks (10 mm diameter) (Croteau and Luoma, 2008; Naylor et al., 1989). The ^{62}Ni was added to both lettuce and leaf disks only in two Ni exposures: Ni-food and Ni-all. Food was soaked for 48 h in separate ^{62}Ni concentrations (1200 $\mu\text{g/L}$ (Ni-food) and 600 $\mu\text{g/L}$ (Ni-all)), gently stirred daily, and

Table 1
Lymnaea stagnalis growth (dry wt.) for all treatments during the 7 d Ni exposures.

Treatments										
Date	Exposure compartment		WD Ref	WD TSS	WD DOC	WD Ni-only	GC Ref	GC TSS	GC DOC	GC Ni-only
08-Apr-10	Ni-water	Mean (mg dry wt)	2.13a	1.37ab	1.67c	0.93ad, cd	1.70 a	0.97ab	1.37c	1.00ad
		St. Dev (mg dry wt)	0.12	0.15	0.15	0.45	0.26	0.29	0.15	0.10
07-May-10	Ni-sediment	Mean (mg dry wt)	2.00a	2.57b	1.73c	1.87d	1.93a	0.00ab	0.00ac	0.00ad
		St. Dev (mg dry wt)	0.17	0.93	0.06	0.32	0.32	0.00	0.00	0.00
27-May-10	Ni-food	Mean (mg dry wt)	2.67a	1.93b	1.73ac	1.70ad	2.57a	1.97b	2.20c	1.87d
		St. Dev (mg dry wt)	0.31	0.40	0.15	0.30	0.32	0.35	0.57	0.81
16-Jun-10	Ni-all compartments	Mean (mg dry wt)	1.10a	0.73b	0.90c	0.77d	1.17a	0.83b	1.10c	0.43ad, cd
		St. Dev (mg dry wt)	0.10	0.32	0.17	0.23	0.32	0.15	0.17	0.06
30-Oct-10	Controls	Mean (mg dry wt)	2.27a	2.60b	2.37c		2.27a	2.07b	2.33c	
		St. Dev (mg dry wt)	0.50	0.44	0.67		0.21	0.32	0.42	

WD = Warden Ditch sediment.

GC = Greenville Creek sediment.

TSS = Total suspended solids.

DOC = Dissolved organic carbon.

Ni = Nickel.

Single letters represent no statistical differences between treatments = a,b,c,d.

Different letters represent statistical differences between treatments = a,b,c,d.

rinsed with Milli-Q water to remove any ^{62}Ni not adsorbed to the food. Leaf disks (dry wt.) and lettuce disks (wet wt.) were weighed prior to ^{62}Ni amendments. All leaf disks were dried at $100 \pm 2^\circ\text{C}$ for 24 h, and lettuce disks air-dried for ~20 min, then wet weighed, and both weighed the nearest 0.01 mg. Leaf/lettuce loss (feeding) was determined by taking the difference from initial (prior to test) food weight minus food weight at test termination.

2.5. Sediment Ni spiking and chemical characterization

For each test, the WDsed and GCsed were spiked with $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, and corrected for dry weight. Sediments were mixed on a mechanized rolling pin for ~2 h, and container headspace was purged of oxygen with N_2 gas. Sediments receiving Ni amendments were held in cold storage ($4 \pm 1^\circ\text{C}$) for 1–3 days. An additional replicate beaker for each treatment was used for sediment chemical characterization, and was separated in two parts: $\text{SEM}_{\text{metal}}/\text{AVS}$, and total metals/TOC. The AVS and SEM_{Ni} were determined following Allen et al. (1991) and Custer et al. (in press). Sediment were microwave digested using Teflon digestion vessels with HNO_3 and HCl for total metals (3:2 ml volume) (USEPA, 2007). All digestates were analyzed by either flame atomic absorption spectrometry or inductively coupled plasma mass spectrometry (ICPMS) for total Ni, Fe, and Mn. Total sediment metal digestions included analytical blanks, procedural blanks, and standards for QA/QC. SEM/AVS analyses included blanks, procedural blanks, spikes, duplicates, and standards. Total organic carbon (TOC) content was determined by following methods described in Heiri et al. (2001). Only Day 7 sediment concentrations (dry weight) are reported, and used in any models of bioavailability (i.e., $\text{SEM}_{\text{Ni}}/\text{AVS}$).

2.6. Organism processing for growth and bioaccumulation

Depending on the test (Ni-water, Ni-sediment, Ni-food, Ni-all) *L. stagnalis* and *H. azteca* survival, growth (dry wt.), ^{62}Ni whole body burden, and leaf/lettuce loss (feeding) were measured. All organisms were counted at test termination (7 d) for survival, growth dry weights (weighed nearest 0.01 mg), and food loss (weighed nearest 0.01 mg). *L. stagnalis* survival was aided with a stereomicroscope to examine movement inside the shell. Leaf and lettuce disks were recovered from the tests; however, at times, *L. stagnalis* consumed entire lettuce disk during the 7 d exposures. All organisms were dried at $100 \pm 2^\circ\text{C}$ for 24 h, and *L. stagnalis* dry weights included shell due to the small size.

In addition, ^{62}Ni whole body burden was determined only in the Ni-food and Ni-all exposures. The use of stable isotopes has many advantages over radioactive isotopes as outlined in Croteau et al. (2004). We chose the stable isotope ^{62}Ni due to its availability (Oak Ridge National Laboratory (Oak Ridge, NC)), low detection with ICPMS, and increased safety (Croteau et al., 2004). After survival was recorded, organisms processed for ^{62}Ni whole body content were soaked in 200 $\mu\text{mol/g}$ of EDTA for 5 min, rinsed with Milli-Q water, dried ($100 \pm 2^\circ\text{C}$ for 24 h), and digested (Hare, 1992; Wilding and Maltby, 2006). Food and organism were digested in acid-cleaned 15 ml centrifuge tubes with 1.3 ml of HNO_3 and 0.2 ml of H_2O_2 , and heated at $65 \pm 5^\circ\text{C}$ for 18 ± 4 h (Solà and Prat, 2006). Due to small size of *L. stagnalis* (<7 d post-hatch), shell and tissue (whole body burden) were digested due to their small size. One of the objectives of this study was to quantify whole body burden, and not to quantify tissue or gut content. Whole body burden metal concentrations can represent realistic ecological food chain effects in natural environmental settings. Hare (1992) discussed the importance of surface bound trace metals in aquatic invertebrates. Cravo et al. (2004) has found that gastropod shells are not good predictors of Ni bioaccumulation. ^{62}Ni in organisms and water

samples was analyzed by ICPMS. Procedural blanks were used during digestions, and for instrument analyses, blanks, standards, and recalibration curves were run every 10–20 samples.

2.7. Data analysis

A one-way ANOVA was used to test for *L. stagnalis* and *H. azteca* survival, growth, or feeding effects with respect to sediment type, treatments, and Ni-compartment (Ni-water, Ni-sediment, Ni-food, Ni-all). Tukey's multiple comparison was used to determine treatment differences when $p \leq 0.05$. Normality and equal variance were tested using Kolmogorov–Smirnov and Levene's test, respectively. A Kruskal–Wallis (KW) non-parametric test was used if assumptions were violated.

The ^{62}Ni whole body burden and subsequent ^{62}Ni :total Ni ratios were tested using the same statistical analyses mentioned above. All ratios were log +1 transformed, and residuals tested for normality and equal variance with the above tests. Bioaccumulation ^{62}Ni :total Ni ratios-ratios were calculated for Ni-water and Ni-food (SI, Table S3), and followed the equations below. All ^{62}Ni and ^{60}Ni ICPMS analyses were corrected for isotopic composition abundances. These values were used to calculate all Ni ratios and ^{62}Ni concentrations (organism, food, and water), and when necessary corrected for organism dry weight. ^{62}Ni -ratios were converted to $\mu\text{g/g}$ or ng/g prior to statistical analyses. All survival, growth, and leaf disk data is presented as mean $\pm$ standard deviation. ^{62}Ni :total Ni ratio for organism and water were calculated as:

$$\text{Ratio (organisms and water)} = \frac{^{62}\text{Ni}}{\text{Total Ni}} \quad (1)$$

where concentrations of ^{62}Ni and ^{60}Ni were determined directly by ICPMS and total Ni was calculated as $^{62}\text{Ni} + 0.96335/0.262231 \times ^{60}\text{Ni}$, with 0.96335 being the natural abundance mole fraction of all Ni stable isotopes other than ^{62}Ni and 0.262231 is the natural abundance mole fraction of ^{60}Ni .

^{62}Ni :total Ni ratio for food was calculated as:

$$\text{Ratio (food)} = \frac{^{62}\text{Ni}}{\text{Total Ni}} \quad (2)$$

where concentrations of ^{62}Ni and ^{60}Ni were determined directly by ICPMS and total Ni was calculated as $^{62}\text{Ni} + 0.96335/0.262231 \times 0.0076/0.9886 \times ^{60}\text{Ni}$, with 0.96335/0.262231 being the same as above for organisms and water and 0.0076/0.9886 is the $^{60}\text{Ni}/^{62}\text{Ni}$ mole ratio in the ^{62}Ni -enriched isotope solution added to food.

3. Results and discussion

3.1. Physicochemical monitoring TSS and DOC

The TSS concentrations were consistent in the controls during the Ni exposures, but the TSS concentrations varied during the TSS treatments (SI, Tables S1 and S2). The mean TSS concentrations for the treatments in all *L. stagnalis* compartment tests were: Reference 0.9 ± 1.3 mg/L, TSS treatments 47.5 ± 16.7 mg/L, DOC treatments 0.9 ± 1.9 mg/L, and Ni-only 3.7 ± 2.9 mg/L (SI, Table S1). The *H. azteca* mean TSS concentrations were: Reference 0.3 ± 0.9 mg/L, TSS treatments 68.1 ± 31.2 mg/L, DOC treatments 1.6 ± 2.3 mg/L, and Ni-only 0.6 ± 1.8 mg/L (SI, Table S2).

The mean DOC concentrations for all *L. stagnalis* compartment tests were: Reference 1.1 ± 0.6 mg/L, TSS treatments 1.5 ± 1.0 mg/L, DOC treatments 9.9 ± 0.3 mg/L, and Ni-only 1.6 ± 0.7 mg/L (SI, Table S1). DOC concentrations in all of the *H. azteca* compartment

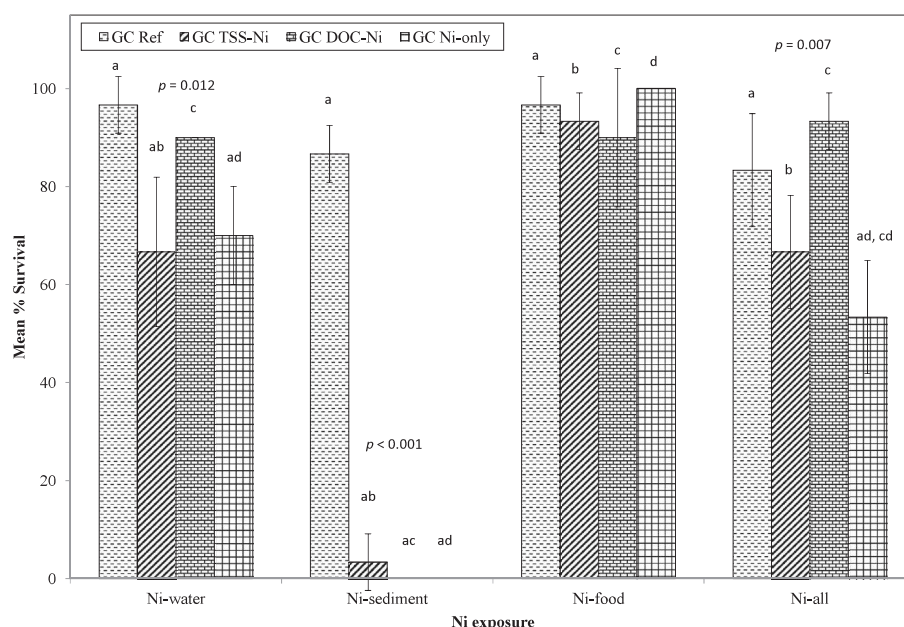


Fig. 1. *Lymnaea stagnalis* survival on GCsed during the Ni exposures. Single letters indicate no statistical difference, and double letters indicate statistical differences between treatments.

tests were: Reference 1.0 ± 0.6 mg/L, TSS treatments 1.3 ± 0.9 mg/L, DOC treatments 10.0 ± 0.6 mg/L, and Ni-only 1.3 ± 0.9 mg/L (SI, Table S2).

These physicochemical parameters are important for characterizing metal toxicity (Pyle et al., 2002; Meyer et al., 2007). Turbidity levels were positively related to increasing TSS concentrations, and Cloran et al. (2010) found this similar linear pattern. The DOC concentrations were adjusted to represent the higher end of DOC concentrations found in natural systems (Wehr et al., 1997). The TSS and DOC concentrations in this study were within the no-effect levels for *Daphnia magna* described in Cloran et al. (2010).

3.2. *L. stagnalis* survival, growth, and feeding effects with sediment type

L. stagnalis showed no survival effects during any of the Ni exposures on WDsed; however, *L. stagnalis* growth was higher in the Ni-DOC (1.67 ± 0.15 mg dry wt.) compared to the Ni-only (0.93 ± 0.45 mg dry wt.) ($p = 0.003$) during the Ni-water exposures (Table 1). During the Ni-food exposures, *L. stagnalis* experienced higher growth in the reference (2.67 ± 0.31 mg dry wt.) than in the Ni-DOC (1.73 ± 0.15 mg dry wt.) and Ni-only (1.70 ± 0.30 mg dry wt.) treatments ($p = 0.015$). The WDsed ^{62}Ni concentrations during the Ni-food exposures ranged from 48.3 to 104.3 $\mu\text{g/g}$ (SI, Table S3).

L. stagnalis experienced increased mortality on GCsed (Fig. 1) during the Ni-water, Ni-sediment, and Ni-all exposures. *L. stagnalis* experienced decreased survival during the Ni-water exposures in both Ni-TSS and Ni-only exposures ($p = 0.012$) (Fig. 1). *L. stagnalis* experienced nearly 100% mortality during the Ni-sediment exposure in all treatments ($p < 0.001$) (Fig. 1). These GC sediment Ni concentrations were between 141 and 164 mg/kg of Ni. This contrasts with <7% mortality in the Ni-sediment exposure (WD sediment) (SI, Fig. S3) at Ni concentrations between 169 and 191 mg/kg. The range of $(\text{SEM}_{\text{Ni-AVS}})/f_{\text{oc}}$ for WDsed was -93.0 to -64.0 $\mu\text{mol/g}$, and 204–323 $\mu\text{mol/g}$ for GCsed (SI, Table S1). During the Ni-all exposure, *L. stagnalis* survival increased (93%) in the DOC treatment versus the Ni-only (53%) ($p = 0.007$). *L. stagnalis* growth

decreased during both the Ni-water, Ni-sediment, and Ni-all exposures on GCsed (Table 1). *L. stagnalis* growth was significantly lower ($p = 0.003$) during the Ni-Water exposure in the Ni-only (1.00 ± 0.10 mg dry wt.) and Ni-TSS (0.97 ± 0.29 mg dry wt.) than the Ni-DOC (1.37 ± 0.15 mg dry wt.) and reference (1.70 ± 0.26 mg dry wt.) (Table 1). During the Ni-all exposure, *L. stagnalis* growth was significantly higher ($p = 0.008$) in the Ni-DOC (1.10 ± 0.17 mg dry wt.) compared to the Ni-only (0.43 ± 0.06 mg dry wt.). The ^{62}Ni food concentrations on GCsed ranged from 57.1 to 97.6 $\mu\text{g/g}$ in the Ni treatments (SI, Table S3).

L. stagnalis experienced decreased survival and growth in response to presence of Ni in multiple compartments and water amendments. The DOC amendments showed attenuated Ni effects on *L. stagnalis* on both WDsed and GCsed during multiple compartment exposures (Ni-water, Ni-all). *L. stagnalis* growth has been shown to be sensitive to Ni by Schlekat et al. (2010) and Brix et al. (2011). Cloran et al. (2010) demonstrated that DOC was able to attenuate *D. magna* Ni toxicity to a certain threshold (<18 mg/L of DOC), and the DOC ranges for the current study were below this concentration. Other studies have shown the importance of DOC as a complexing ligand for metals, thus reducing bioavailability, and any subsequent toxicity (Playle, 2004; Di Toro et al., 2001). In addition to DOC, nickel bioavailability appears to have been reduced on WD sediment exposures. These sediments had higher in AVS and TOC, and potentially have more ligands available for Ni complexation. The differences between the $(\text{SEM}_{\text{Ni-AVS}})/f_{\text{oc}}$ are demonstrating Ni bioavailability increased on GCsed, and contributed to low survival of *L. stagnalis* during these 7 d Ni-sediment exposures. This range of the $(\text{SEM}_{\text{Ni-AVS}})/f_{\text{oc}}$ (-243.4 to -64.0 $\mu\text{mol/g}$ of OC) model was below the theoretical no-effect threshold (<130 $\mu\text{mol/g}$ of OC) for WD sediment, and the $(\text{SEM}_{\text{Ni-AVS}})/f_{\text{oc}}$ values for GC sediments (121.0–323.4 $\mu\text{mol/g}$ of OC) fell either slightly below or within the area of uncertainty (USEPA, 2005). Of importance is the increased feeding *L. stagnalis* demonstrated during the Ni-food and Ni-all exposures regardless of sediment type (SI, Table S4). This increased feeding occurred during the ^{62}Ni -labeled exposures (Ni-Food and Ni-all) (SI, Table S4), and similar results have been observed in other studies by Croteau and

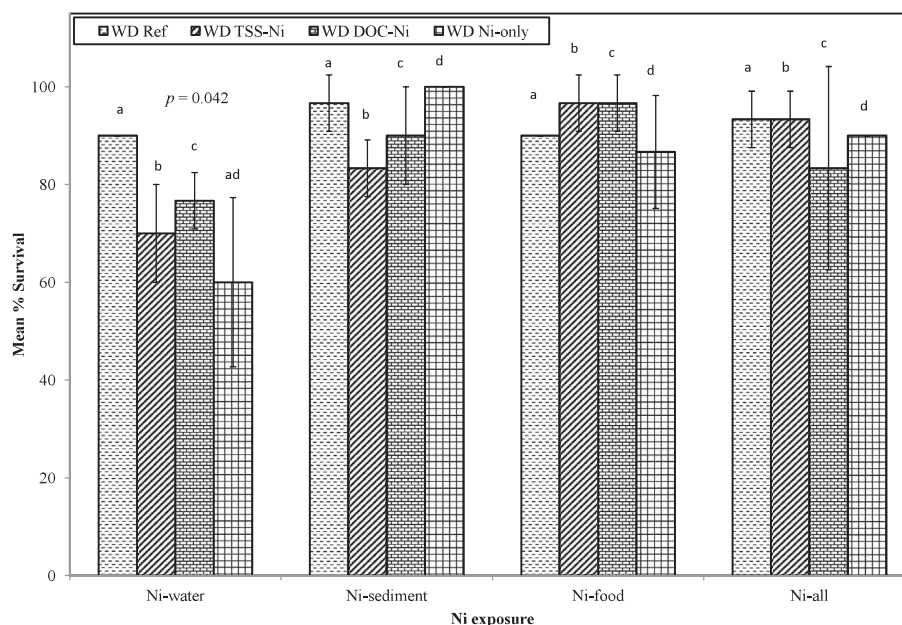


Fig. 2. *Hyalella azteca* survival on WDsed during the Ni exposures. Single letters indicate no statistical difference, and double letters indicate statistical differences between treatments.

Table 2

Hyalella azteca growth (dry wt.) for all treatments during the 7 d Ni exposures.

Treatments										
Date	Exposure compartment		WD Ref	WD TSS	WD DOC	WD Ni-only	GC Ref	GC TSS	GC DOC	GC Ni-only
16-Apr-10	Ni-water	Mean (mg dry wt)	0.73a	0.53b	0.37ac	0.10 ad, bd	0.97 a	0.43ab	0.77c	0.40 ad
		St. Dev (mg dry wt)	0.06	0.21	0.12	0.10	0.38	0.15	0.25	0.10
18-May-10	Ni-sediment	Mean (mg dry wt)	0.87a	0.17ab	0.70bc	0.40d	0.83a	0.20b	0.10ac	0.10ad
		St. Dev (mg dry wt)	0.25	0.15	0.20	0.17	0.42	0.17	0.10	0.17
07-Jun-10	Ni-food	Mean (mg dry wt)	0.57a	0.50b	0.80bc	0.63d	0.63a	0.70b	0.43c	0.63d
		St. Dev (mg dry wt)	0.12	0.10	0.00	0.15	0.06	0.30	0.15	0.32
25-Jun-10	Ni-all compartments	Mean (mg dry wt)	0.80a	0.57b	0.80c	0.67d	0.93a	0.47b	0.43c	0.17ad
		St. Dev (mg dry wt)	0.10	0.31	0.12	0.15	0.06	0.38	0.21	0.29
25-Oct-10	Controls	Mean (mg dry wt)	2.20a	1.73b	1.83c		1.67a	1.50b	1.63c	
		St. Dev (mg dry wt)	0.40	0.38	0.31		0.31	0.30	0.45	

WD = Warden Ditch sediment.

GC = Greenville Creek sediment.

TSS = Total suspended solids.

DOC = Dissolved organic carbon.

Ni = Nickel.

Single letters represent no statistical differences between treatments = a,b,c,d.

Different letters represent statistical differences between treatments = a,b,c,d.

Luoma (2008, 2009). Possible explanation for increased feeding in our study could be either the lack of (Ni-Food) or lowered (Ni-all) Ni concentrations in the other compartments (water and sediment). These Ni-water and Ni-sediment tests did not show these same feeding effects (SI, Table S4).

3.3. *H. azteca* survival, growth, and feeding effects and sediment type

H. azteca survival was sensitive to waterborne Ni (Ni-water) (Fig. 2) on WDsed; however, no other survival effects were detected in any of the Ni compartment exposures. *H. azteca* survival during the Ni-water exposure decreased ($p = 0.042$) from $90 \pm 0\%$ in the Reference treatment to $60 \pm 12\%$ in the Ni-only treatment. *H. azteca* growth effects were observed during the Ni-water, Ni-food, and Ni-sediment exposures on WDsed. During the Ni-water exposures, *H. azteca* growth was higher ($p = 0.002$) in the TSS treatments

(0.53 ± 0.21 mg dry wt.) compared to the Ni-only (0.10 ± 0.10 mg dry wt.). Contrastingly, during the Ni-sediment exposure, *H. azteca* growth was lower ($p = 0.011$) in the TSS (0.17 ± 0.15 mg dry wt.) and higher in the Ni-DOC (0.70 ± 0.20 mg dry wt.) and the reference (0.87 ± 0.25 mg dry wt.) treatments (Table 2). A similar effect was observed in the Ni-food exposure with higher ($p = 0.045$) growth in the Ni-DOC (0.80 ± 0.00 mg dry wt.) compared to the TSS (0.50 ± 0.10 mg dry wt.) treatment. During the Ni-water and Ni-all, *H. azteca* leaf disk processing was greater in the reference treatment and significantly less feeding in all of the Ni treatments (SI, Table S5).

On the GCsed, *H. azteca* survival effects were observed during most Ni compartment tests (Ni-water, Ni-sediment, and Ni-all exposures) (Fig. 3). During the Ni-water exposure, *H. azteca* survival was lower in the TSS ($55 \pm 10\%$), but improved in the Ni-DOC ($77 \pm 6\%$) and Ni-only treatment ($80 \pm 22\%$) ($p = 0.008$). Regardless of water amendments (TSS, DOC, or none), *H. azteca* survival in

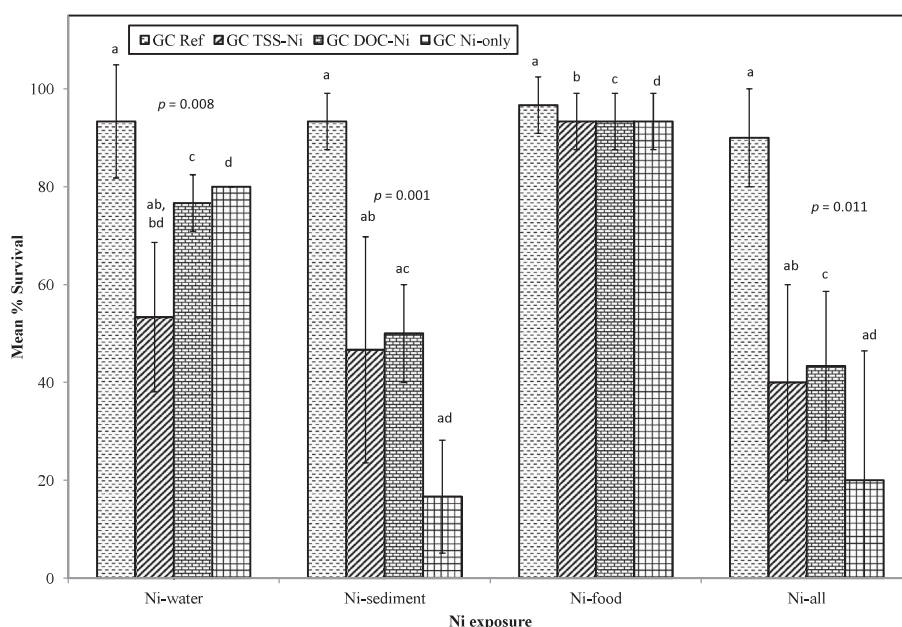


Fig. 3. *Hyalella azteca* survival on GCsed during the Ni exposures. Single letters indicate no statistical difference, and double letters indicate statistical difference between treatments.

Ni-sediment ($p = 0.001$) and Ni-all ($p = 0.011$) exposures showed strong responses with decreasing survival in the presence of Ni (Fig. 3). *H. azteca* experienced similar growth effects on the GCsed as were observed on WDsed during the Ni-water and Ni-sediment. However, no Ni-food effects were observed, but *H. azteca* did experience growth effects in the Ni-all exposures (Table 2). *H. azteca* growth decreased ($p = 0.018$) during the Ni-sediment in the DOC and Ni-only treatments on GCsed. In the Ni-all exposure, *H. azteca* growth was lowest in Ni-only (0.17 ± 0.29 mg dry wt.) and higher in the reference (0.93 ± 0.06 mg dry wt.). *H. azteca* experienced reduced feeding during the Ni-all exposure in all treatments ($p = 0.020$) on GCsed (SI, Table S5). *H. azteca* feeding was lowest in the Ni-only treatment (0.00 ± 0.00 mg dry wt.) when compared to the reference (-0.57 ± 0.35 mg dry wt.) (SI, Table S5).

DOC, TSS, and sediment type are important factors for attenuating Ni effects on *H. azteca*. The TSS amendments showed both ameliorative Ni effects, and enhanced Ni effects when compared to the DOC treatments. The TSS protective effects were observed during the Ni-water exposures (WDsed), and TSS enhanced effects were observed during the Ni-sediment (WDsed) and Ni-food (WDsed) exposures, both with WD sediment. These differences between TSS and DOC treatments are showing significantly higher growth in the DOC amendments during the Ni-sediment and Ni-food exposures on WDsed. However, these DOC growth improvements were only higher when compared to the TSS exposures, and did not follow the defined DOC protective effect: DOC endpoint > Ni-only endpoint. Doig and Liber (2006) found that DOC affected Ni tissue accumulation at low Ni concentration, but did not improve *H. azteca* survival regardless of DOC concentration. The differing results above are similar to the Ni-TSS reduced effects observed in Pyle et al. (2002); however, Cloran et al. (2010) showed Ni-TSS toxicity to a threshold.

Sediment type appears to affect *H. azteca* survival in the presence of Ni. *H. azteca* survival and growth was lowest during the Ni-sediment and Ni-all exposures on GCsed when compared to WDsed (Figs. 2–3). These WDsed may be contributing to reduced Ni bioavailability through complexation and adsorption (Mahony et al., 1996; USEPA, 2005; Di Toro et al., 2005). The ranges of

(SEM_{Ni-AVS})/*foc* in the Ni-all exposure for WDsed were -172 to -139 $\mu\text{mol/g}$ OC, and in GCsed, the range was from 104 to 187 $\mu\text{mol/g}$ OC. Both sediment types were at or below the predicted no-effect threshold levels of the (SEM_{Ni-AVS})/*foc* model (USEPA, 2005). These WDsed are higher in AVS, TOC, and Fe than GCsed, and in addition to increased ligand availability, they may offer greater nutritional value to organisms.

H. azteca feeding inhibitions were observed in the Ni-water and Ni-all exposures on WDsed and in the Ni-all GCsed. These results suggest that food was avoided when Ni was amended to water in both the Ni-water and Ni-all exposures. *H. azteca* fed less during the Ni-water and Ni-all exposures, and their growth was low during these exposures (Table 2 and SI, Table S5). Overall, *H. azteca* growth was lowest during the Ni-sediment (GCsed), but this exposure saw no decrease in feeding between treatments. During Ni waterborne exposures, Doig and Liber (2006) suggested that Ni is complexing with food and causing avoidance, and our results suggest similar effects in this study. Nguyen et al. (2012) showed similar food avoidance and decreased growth in *H. azteca* when exposed to both Zn amended food and sediments. Hatch and Burton (1999) and Gillespie et al. (1997) have shown that *H. azteca* leaf processing was negatively affected with increasing chemical concentrations.

3.4. *L. stagnalis* ^{62}Ni transfer during Ni-food and Ni-all exposures

Nickel was accumulating in *L. stagnalis* during the Ni-food and Ni-all exposures, with water amendments and sediment type attenuating accumulation. Differences in *L. stagnalis* ratio ^{62}Ni organism:Total Ni Water during the Ni-food exposures (WDsed) were observed (TSS (18051), DOC (16,321), Ni-only (5389), reference (2830)), but were not significantly different ($p = 0.069$). *L. stagnalis* had similar ^{62}Ni whole-body concentrations among the Ni treatments during the Ni-food exposures: TSS (7.6 ± 4.0 $\mu\text{g/g}$), DOC (8.0 ± 4.9 $\mu\text{g/g}$), and Ni-only (6.5 ± 2.5 $\mu\text{g/g}$) treatments on WDsed (SI, Table S3). The ^{62}Ni water column concentrations ranged from 0 to 0.25 $\mu\text{g/L}$ during the Ni-food exposures. During the Ni-all exposures (WD sediment), *L. stagnalis* ^{62}Ni whole body burden concentrations increased ($p = 0.041$) in the TSS treatment (7.3 ± 4.6 $\mu\text{g/}$

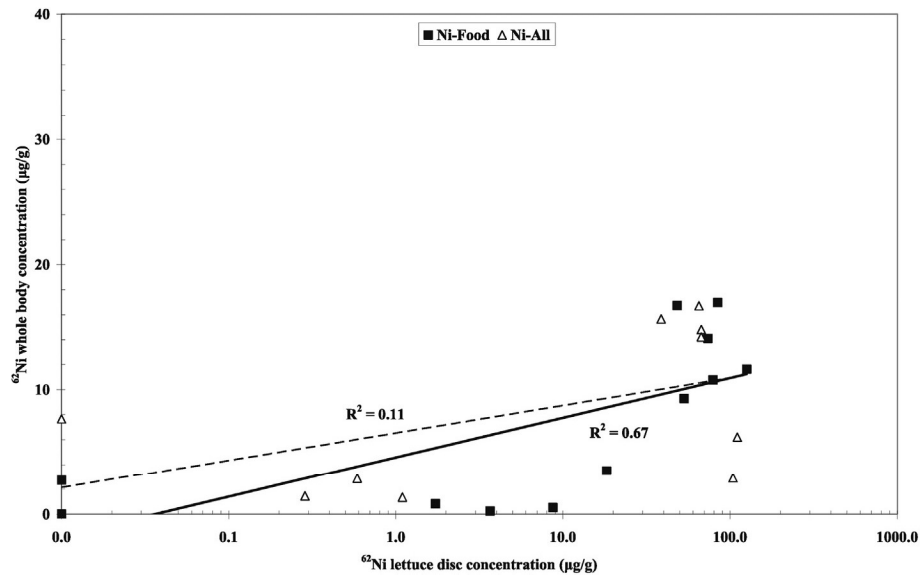


Fig. 4. *Lymnaea stagnalis* relationship between ^{62}Ni whole-body concentrations and ^{62}Ni lettuce concentrations on GCsed. Solid line is Ni-food test, and dashed line is Ni-all test.

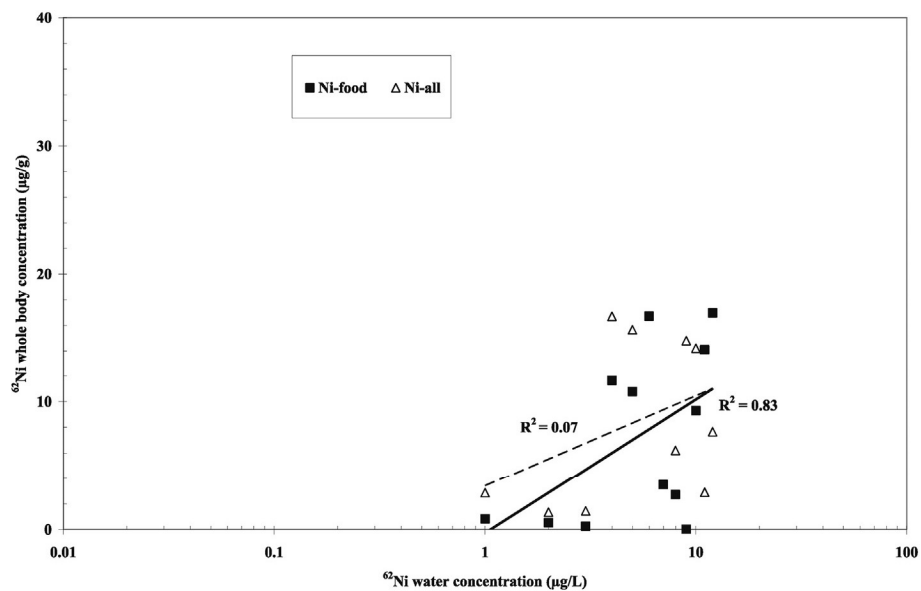


Fig. 5. *Lymnaea stagnalis* relationship between ^{62}Ni whole-body concentrations and ^{62}Ni water concentrations on GCsed. Solid line is Ni-food test, and dashed line is Ni-all test.

g) compared to the reference ($0.2 \pm 0.3 \mu\text{g/g}$). The ^{62}Ni water column concentrations ranged from 6.0 to 10.6 $\mu\text{g/L}$ during the Ni-all exposures. There were no other effects observed between any of the ^{62}Ni ratios on WDsed during the Ni-all exposure.

The *L. stagnalis* ^{62}Ni whole-body concentrations on GCsed during the Ni-food exposure was significantly different ($p < 0.001$): Reference ($0.5 \pm 0.3 \mu\text{g/g}$), DOC ($2.1 \pm 1.8 \mu\text{g/g}$), TSS ($13.0 \pm 3.2 \mu\text{g/g}$), and Ni-only ($13.5 \pm 3.9 \mu\text{g/g}$) (SI, Table S3). The *L. stagnalis* ^{62}Ni organism:total Ni Food ratio on GCsed was significantly lower ($p = 0.021$) with the DOC (0.00) treatment being lower than the Ni-only (0.07) treatment. As observed during the Ni-food (WD sediment) exposures, ^{62}Ni water column concentrations (0.05–0.44 $\mu\text{g/L}$) were detected in the GCsed treatments. The ^{62}Ni organism:total Ni Water ratio on GCsed showed lower ($p = 0.001$) values in the Reference (1462) and DOC (2661) treatments and higher in the Ni-only (15,151) and TSS (17,586) treatments during the Ni-food

exposure. All of these measures are suggesting ^{62}Ni protective effects of DOC. During the Ni-all exposure (GCsed), *L. stagnalis* ^{62}Ni whole-body concentrations were significantly higher ($p = 0.045$) in the TSS ($16.2 \pm 0.7 \mu\text{g/g}$) exposures compared to the DOC ($10.5 \pm 6.1 \mu\text{g/g}$) and Ni-only ($8.3 \pm 5.7 \mu\text{g/g}$) exposures. *L. stagnalis* whole-body ^{62}Ni concentrations increase with ^{62}Ni in food ($p = 0.001$, $R^2 = 0.67$) and ^{62}Ni in water ($p < 0.001$, $R^2 = 0.83$) (Figs. 4–5). The ^{62}Ni water concentrations ranged from 2.28 to 9.47 $\mu\text{g/L}$ during the Ni-all exposures. *L. stagnalis* ^{62}Ni Bio-concentration factor (BCF) (^{62}Ni organism/ ^{62}Ni water) for both sediment types (WD and GC) during the Ni-all exposure was inversely related to ^{62}Ni water concentration; however, in the Ni-food exposure the slope appears to approach 0 (SI, Fig. S2).

Several factors are contributing to, and controlling ^{62}Ni bio-accumulation during the Ni-food and Ni-all exposures. Unexpectedly, ^{62}Ni fluxed from food to water column and contributed to low

^{62}Ni concentrations. This is supported by the ^{62}Ni organism:total Ni Water ratios showing increased ^{62}Ni concentration associated with solids (TSS) and DOC; however, *L. stagnalis* whole body ^{62}Ni concentrations were not different among the treatments on WDsed in the Ni-food exposures. Contrastingly, on GCsed, the TSS and Ni-only treatments showed increased ^{62}Ni *L. stagnalis* bioaccumulation, and DOC attenuated ^{62}Ni bioaccumulation. These data suggest DOC provides a protective effect from ^{62}Ni bioaccumulating in *L. stagnalis*. Croteau and Luoma (2008) found higher ^{62}Ni tissue concentrations in *L. stagnalis* from food versus water, and Nguyen et al. (2012) found that Zn amended food was not contributing to an increase in overlying water Zn concentrations. The results from our study suggest that it may be a combination of ^{62}Ni water and ^{62}Ni food concentrations contributing to higher ^{62}Ni body burden rather than diet alone. Differences between the studies could be attributed to ^{62}Ni flux from food to water (current study), and endpoint selection. Croteau and Luoma (2008) examined tissue concentration, and our study utilized whole-body concentrations (including gut contents and shell). Doig and Liber (2006) found strong correlations between increasing Ni^{2+} and Ni-tissue concentrations. They also found DOC effects at low Ni concentrations, and our results show a similar trend. During the Ni-all exposures, *L. stagnalis* were bioaccumulating more ^{62}Ni in the TSS and Ni-only treatments regardless of sediment type. This is suggesting that TSS had a synergistic effect on ^{62}Ni accumulation regardless of sediment type. These TSS ^{62}Ni body burden results (Ni-all) are similar to the TSS results (Ni-food) which showed an increase of ^{62}Ni *L. stagnalis* body burden. Unlike the results in the Ni-food, the Ni-all exposures saw no DOC mitigating effect. DeForest et al. (2007) suggested that *L. stagnalis* BAFs (water and food) may possibly bioaccumulate copper given their non-negative BAF slopes. The current study is showing less pronounced negative slope (slope closer to 0) with Ni-Food BCF (SI, Fig. S2). Additionally, *L. stagnalis* ^{62}Ni whole-body concentrations increased with increasing ^{62}Ni food and ^{62}Ni water (Figs. 4–5). These results suggest that when Ni is present in multiple compartments (water and food), *L. stagnalis* may bioaccumulate Ni, and these results represent realistic conditions that occur in natural systems.

3.5. *H. azteca* ^{62}Ni transfer in Ni-food and Ni-all exposures

Suspended solids can play an important role in Ni bioavailability and Ni bioaccumulation. During the Ni-food exposures (WDsed), *H. azteca* ^{62}Ni whole-body concentrations were significantly higher ($p = 0.009$) in the Ni-TSS ($9.8 \pm 4.6 \mu\text{g/g}$) compared to the reference ($1.0 \pm 0.6 \mu\text{g/g}$), DOC ($1.9 \pm 0.7 \mu\text{g/g}$), and Ni-only ($2.3 \pm 1.9 \mu\text{g/g}$) (SI, Table S6). A similar result was observed with *H. azteca* ^{62}Ni organism:Total Ni Water ratio (Ni-food) showing higher TSS (39,174) results compared to DOC (2924)/Ni-only (3738), but was not significant (KW $p = 0.099$) (SI, Table S6). The BCF vs. ^{62}Ni water showed a negative linear response with decreasing BCFs to increasing ^{62}Ni water concentrations (SI, Fig. S2). *H. azteca* showed no treatment differences and little ^{62}Ni bioaccumulation during the Ni-food exposures on the GC sediment. During the Ni-all exposures, *H. azteca* ^{62}Ni whole-body concentrations showed no differences on either WD or GCsed. The ratios of ^{62}Ni organism:total Ni Water (Ni-all) were significant ($p = 0.002$) different among treatments on WDsed, but no treatment effects were detected: DOC (28), TSS (6), and Ni-only (14). Similar results showed the ratio of ^{62}Ni organism:total Ni Water (Ni-all) on GCsed was significant, and but no treatment effects: DOC (14), TSS (20), and Ni-only (12).

H. azteca whole-body ^{62}Ni concentrations were significantly higher in the TSS treatment on WDsed. Weak relationships were observed between ^{62}Ni food and ^{62}Ni whole-body concentration. The TSS effects are suggesting ^{62}Ni fluxed from food as seen in the

L. stagnalis Ni-food tests; and provided an additional route of exposure (^{62}Ni adsorbing to TSS particles) as observed from ^{62}Ni organism:total Ni Water ratios (Table S6). Pyle et al. (2002) demonstrated that TSS was ameliorating the effects of Ni on *D. magna*, but Cloran et al. (2010) found the Ni-TSS exposures were contributing to toxicity in *D. magna* above a certain threshold. During this study, there was a visible layer of TSS particles settling out on the sediments and leaf disks after 24 h. These ^{62}Ni whole-body concentrations increases in *H. azteca* could be explained by their shredding behavior, and subsequent indirect consumption of TSS- ^{62}Ni adsorbed particulate material while feeding.

4. Conclusions

The results from our study suggest that Ni-compartmental differences are essential to understanding Ni toxicity to *H. azteca* and *L. stagnalis*. Whether Ni is amended to water, sediment, food, or in combination, these exposures represent realistic conditions encountered by aquatic organisms in natural systems. The objectives of this study were to discern Ni toxic effects on *H. azteca* and *L. stagnalis* Ni amendments to water, sediment, and food, with varying overlying water quality (TSS and DOC) on different sediment types. Waterborne Ni appeared to drive the survival and growth effects in both organisms, followed by sediment, and then food. Previous studies (Custer et al. in review; Costello et al., 2011, 2012) have demonstrated that sediment type is an important determinant controlling Ni bioavailability and limited ^{62}Ni bioaccumulation in this study. The DOC and TSS amendments were protective of Ni toxicity; however, TSS also enhanced Ni toxicity and ^{62}Ni bioaccumulation. Across many levels of biological responses (survival, growth, whole-body concentration), DOC amendments provided protective effects from Ni on *L. stagnalis*.

Direct transfer of ^{62}Ni from food was negligible, and complicated by the unexpected desorption of ^{62}Ni from food to the water column. Diet (^{62}Ni -food) alone was not contributing to adverse survival or growth effects in the *L. stagnalis* and *H. azteca*. These results suggest that ^{62}Ni desorbing from food to water column was adsorbed with TSS and complexed with DOC. Subsequently, both *L. stagnalis* and *H. azteca* showed increasing ^{62}Ni bioaccumulation with TSS amendments. However, during the Ni-all exposure, *L. stagnalis* bioaccumulation increase may have been a function of multiple Ni exposures from the other compartments (water, food, and sediment). Previous research has shown that *H. azteca* and *L. stagnalis* have accumulated Ni during water-only and food studies (Doig and Liber, 2006; Croteau and Luoma, 2008). Other studies have shown that metals associated with food do bioaccumulate (Courtney and Clements, 2002; Wilding and Maltby, 2006), and some results have shown that *L. stagnalis* metal tissue content increased during feeding rather than water-only acute metal exposures (Croteau and Luoma, 2008, 2009). *L. stagnalis* was more sensitive than *H. azteca*, and Schlegel et al. (2010) and Brix et al. (2011) have shown that *L. stagnalis* is a sensitive organism to Ni and Cu, respectively. This research has demonstrated the importance of Ni toxicity and bioaccumulation, and may contribute to water quality criteria and sediment quality guidelines for assisting regulatory decisions regarding Ni in aquatic environments.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.envpol.2015.09.045>.

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